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TITLE PAGE

Problem Statement ID - **SIH26253**

Problem Statement Title - **Development of Strip-Based Water Quality Test Kits for Assessment of Free Residual Chlorine (FRC) with Sensitivity up to 0.1 ppm for Field-Level Applications**

Theme - **Clean & Green Technology**

PS Category - **Hardware**

Team ID - **151508**

Team Name (Registered on portal) - **PARAKRAM**



PARAKRAM

JalSatya (जलसत्य)
A chlorine test with proof built in

No strip, no record. No guesswork at the tap.

PARAKRAM

EVERY TEST LEAVES PROOF

A dye-free strip designed to read free chlorine to 0.1 ppm, with proof of every test

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Proposed Solution (Describe your Idea/Solution/Prototype)

JJM's 2021 framework [1]: "the availability and usage of these kits is unknown."
CAG audit, 2026 [2]: more tests shown than samples received in four Thane villages.
With JalSatya, no strip means no record, and every record carries its proof.

Detailed explanation of the proposed solution

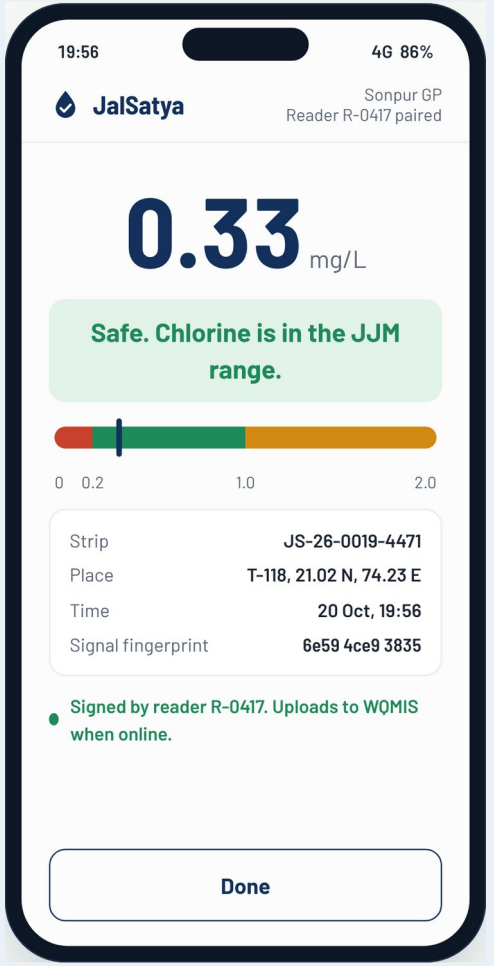
- Dry printed-carbon strip with a buffer pad. No colour dye.
- A pocket reader turns its current into mg/L, about 10 s from dip to result.
- Each result is signed with strip serial, GPS and time, ready for WQMIS.

How it addresses the problem

- Nothing to judge by eye [5]: one button, Hindi voice, works offline.
- One serial per strip, so tests reported can never exceed strips issued.

Innovation and uniqueness of the solution

- Proof-of-Test: a serialised, signed strip. No real electrode curve, no record.
- A village twin turns verified readings into the right dose, and flags readings that do not fit the tap.



Tried in our demo, and refused

Typed number
Refused: no electrode signal

Same strip twice
Refused: serial already spent

Copied curve
Refused: fingerprint seen before

Invented curve
Refused: unsigned, wrong physics

Simulated: all 2,000 fakes across the 5 attacks we modelled were refused; 99.3% of real tests were accepted.

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TECHNICAL APPROACH

Measured by electrochemistry, not by eye, and signed before it leaves the reader

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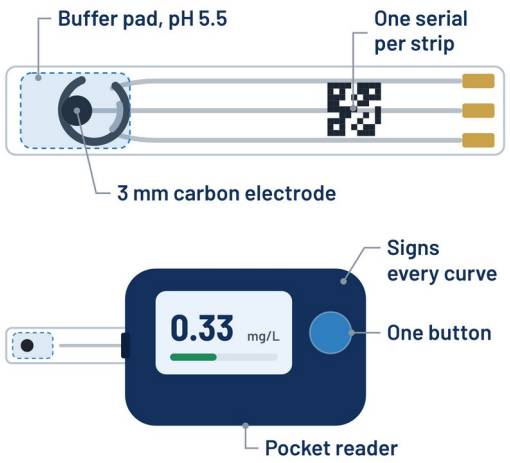
Technologies to be used (e.g. programming languages, frameworks, hardware)

- Screen-printed carbon electrode strip
- Dried buffer pad, pH 5.5
- Potentiostat front end
- ESP32 reader, C firmware
- BLE and USB-C
- Android app in Kotlin, offline first
- Ed25519 signed records
- WQMIS sync
- Python physics model
- Web dashboard, ECharts

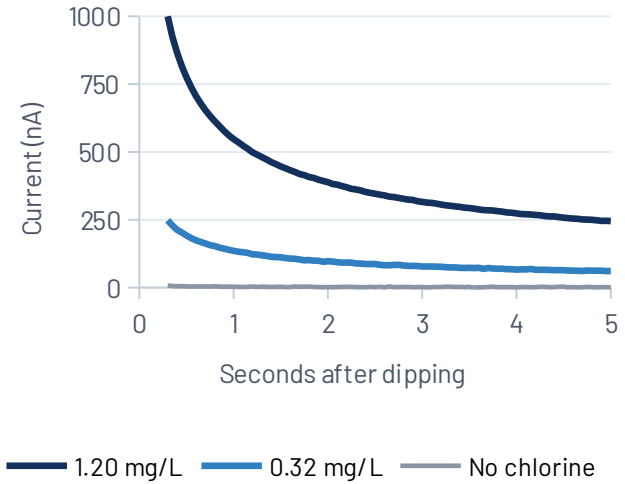
Methodology and process for implementation (Flow Charts/Images/ working prototype)

1. **Scan** — Strip serial and expiry checked, assigned tap shown
2. **Dip and read** — 3 s dip, then a 5 s current reading
3. **Verify** — One-time strip, reader signature, curve shape
4. **Sync** — Signed record to WQMIS, queued when offline
5. **Advise** — Twin updates the dose and the next taps to test

Strip and reader (design)



Prototype strip current (simulated)



Why the reading is true

Current falls as Cottrell's law says

$$i = nFAC\sqrt{(D / \pi t)}$$

About 43 nA for every 0.1 mg/L at 1 s on a 3 mm electrode.

Share of chlorine the electrode sees, as HOCl

$$f = 1 / (1 + 10^{(pH - 7.54)})$$

99% at the pad's pH 5.5, for water from pH 6 to 9.

Why the record is real

One strip, one test
the serial is spent on first use

Signed at the source
the reader signs the raw curve, place and time

Physics check
a real curve falls as $1/\sqrt{t}$; every fake shape we modelled failed

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FEASIBILITY AND VIABILITY

Method, maths and software are built. The working strip comes to the finale.

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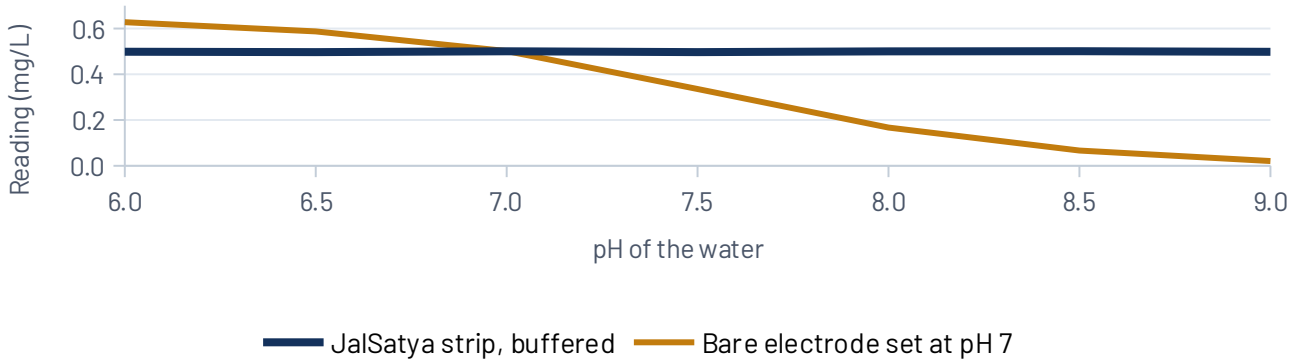
Analysis of the feasibility of the idea

0.027 mg/L detection limit, under 0.1 ppm	**98%** of readings within ±0.1 mg/L	**10 s** from dip to a signed result	**100%** of 2,000 simulated fakes refused

Simulated [12] with real-world noise and strip spread. 99.3% of real tests pass.

The same 0.50 mg/L water, read at different pH (simulated)

At pH 8.5 a bare electrode set at pH 7 shows only **13% of the chlorine**. The buffered strip still reads 0.50.



Fits rules already in force. The JJM framework allows instruments like a chlorimeter and asks for an FTK inventory with reagent expiry dates [1, Sec 4.3]. JalSatya is a strip chlorimeter that keeps its own inventory.

Potential challenges and risks

Strategies for overcoming these challenges

Chloramine or manganese	Low working potential; lab checks with IS 3025 and DPD [13]
Strip spread, heat and humidity	Batch calibration code; dry strip sealed in a foil pouch
Wrong water or tampered reader	Twin flags misfit readings; secure key chip; every fake burns a strip
Testers new to a device	One button, Hindi voice prompts, the same trained women
Cost at national scale	Target under Rs 10 a strip at volume, under Rs 2,500 a reader (our estimate)

Roadmap

Today → App, reader logic, physics model, twin, dashboard

Finale → Working strip and reader, checked against DPD

Pilot → One block with a state water department

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IMPACT AND BENEFITS

For 24.80 lakh testers, every pump operator and 15.72 crore homes on tap water

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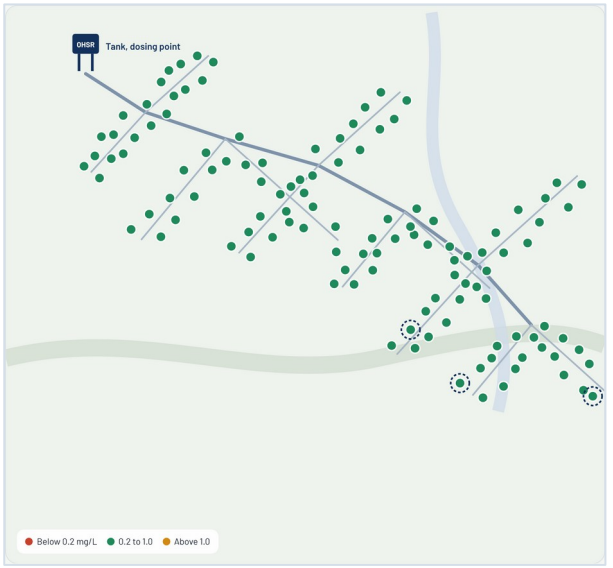
Potential impact on the target audience

Village chlorine twin on a sample scheme of 120 taps (simulated)

Each dot is a household tap. **Red** below 0.2 mg/L, **green** 0.2 to 1.0, **amber** above 1.0 [4]



Usual dose 1.00 mg/L
108 of 120 taps safe. **Far end 0.18 mg/L**



Twin's dose 1.26 mg/L
120 of 120 taps safe. **Far end 0.26 mg/L**

Village testers	A 10-second test, no colour to judge, Hindi voice
Pump operators	The exact dose to add, and an alert when a far tap runs low
Block and district	Only verified tests, with kit stock and expiry tracked
DDWS and NJJM	WQMIS numbers that stand up to audit

Benefits of the solution (social, economic, environmental, etc.)

Social safe chlorine at the far end of every line

Economic no dye reagent to buy or throw away

Environmental no o-tolidine, no liquid waste in villages

Governance test counts that match strips issued

≥1.2 crore
tests a year: 2 tail ends × 5.07 lakh villages × 12

24.80 lakh
women already trained to test with field kits [3]

5.07 lakh
villages where those trained testers work [3]

Larger vision. pH, fluoride and nitrate strips on the same reader, a live national chlorine map, and links to JJM tank sensors. States buy readers once and strips from WQMS funds.

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RESEARCH AND REFERENCES

Every claim here traces to a public source or to our simulation

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Details / Links of the reference and research work

Problem and policy

[1] DDWS, Ministry of Jal Shakti, Drinking Water Quality Monitoring and Surveillance Framework, 2021. jaljeevanmission.gov.in

[2] CAG, Performance Audit on Jal Jeevan Mission (Maharashtra), Report No. 2 of 2026, Chapter V. cag.gov.in

[3] PIB, Jal Jeevan Mission factsheet, data as on 21 and 22 October 2025. pib.gov.in

[4] BIS, IS 10500:2012, Drinking Water Specification (free residual chlorine 0.2 to 1 mg/L). consumerhelpline.gov.in

[5] Raj and Jamwal, How reliable are water quality monitoring results from field test kits, Down To Earth, Dec 2022. downtoearth.org.in

Existing approaches

[6] González Gómez et al., Color QR Codes for Smartphone-Based Analysis of Free Chlorine in Drinking Water, Sensors, 2025. doi:10.3390/s25113251

[7] Palintest ChloroSense, reagent-free electrochemical chlorine meter, 2008. wateronline.com

[8] Impact of gate electrode on free chlorine sensing performance in solution-gated graphene field-effect transistors, RSC Advances, 2024. doi:10.1039/D3RA07692J

[9] Shekhar et al., A Disposable Microsensor for Continuous Monitoring of Free Chlorine in Water, IEEE Sensors, 2003. pure.korea.ac.kr

Science and test methods we build on

[10] Bard and Faulkner, Electrochemical Methods, 2nd ed., Wiley, 2001 (Cottrell equation)

[11] Morris, J. C., The acid ionization constant of HOCl from 5 to 35 °C, J. Phys. Chem. 70, 3798, 1966

[12] Team PARAKRAM, JalSatya simulation and demo. github.com/codedbysuryansh/PARAKRAM-SIH26253

[13] BIS, IS 3025 (Part 26), Methods of Sampling and Test for Water and Wastewater: Chlorine, Residual

What exists, and what is new

DPD and OTO colour kits [5]	judged by eye; reagents expire
Phone colour apps [6]	qualitative, sensitive to light
Electrochemical meters [7, 8, 9]	precise, but costly and no audit trail
JalSatya	**serialised, signed strip, plus dose advice for the village**

See it run

Demo codedbysuryansh.github.io/PARAKRAM-SIH26253
Code github.com/codedbysuryansh/PARAKRAM-SIH26253
Video youtu.be/rtyKnEUzjjI

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No strip, no record. No guesswork at the tap.